

ADITYA RAINA

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PROFESSIONAL SUMMARY

Software Engineer with hands-on experience designing and building full-stack applications, distributed systems, DevOps pipelines, and cross-chain blockchain protocols. Proficient in C/C++, Python, TypeScript, Go, and Solidity, with a strong track record of delivering production-grade tools, including a security scanning platform published on PyPI and GitHub Marketplace. Skilled in cloud infrastructure, CI/CD automation, network systems, and machine learning, with a focus on building scalable, reliable, and security-conscious software.

TECHNICAL SKILLS

Languages: Go, C/C++, Python, Java, Rust, TypeScript, JavaScript, Solidity

Backend & Web: Next.js, React, Node.js, FastAPI, Django, Express.js, REST APIs

Cloud & DevOps: Docker, Kubernetes, Jenkins, Argo CD, GitHub Actions, GitLab CI/CD, Terraform, Ansible, Prometheus, Grafana

Systems & Security: Distributed Systems, Network Security, DevSecOps, Smart Contract Security, Fuzzing

Blockchain / Web3: Ethereum (ERC-20/721), Solana (Anchor), Foundry, Wormhole, Noir (zk), Slither

EXPERIENCE

DevOps Intern — IBM

May 2026 – Jul 2026

- Built and maintained CI/CD pipelines using Jenkins, GitHub Actions, and containerized services with Docker/Kubernetes to automate and streamline deployment workflows.
- Provisioned cloud infrastructure with Terraform and Ansible, and configured Grafana and Nagios dashboards for monitoring system health and pipeline performance.

AI/ML Intern — Coratia Technologies

Dec 2023 – Feb 2024

- Built an NLP-based sentiment analysis system in Python, classifying user sentiment across large-scale text datasets using supervised machine learning models.
- Engineered preprocessing pipelines (tokenization, vectorization) and tuned model evaluation metrics, improving classification accuracy.

PROJECTS

SecureOps — Contextual DevSecOps Security Platform

- Built a Go/Chi DevSecOps platform orchestrating 6 scanners (Gitleaks, Syft, Gype, Semgrep, Trivy, OWASP ZAP) behind one adapter interface, normalizing their output into a canonical finding model keyed by a collision-resistant SHA-256 fingerprint, correlating findings across security domains into contextual issues, and scoring them with a deterministic 5-factor risk engine
- Shipped it four ways from one codebase — REST API, Next.js dashboard, CI client with a three-value exit contract, and a GitHub Action — hardened with Argon2id accounts, project-scoped RBAC, transactional append-only audit logging, and a Helm chart that refuses any image not pinned by digest.

ChainAudit — Smart Contract Security Platform

- Built a multi-chain smart contract security scanner mapping 51 Slither detectors to 30 CVSS-scored rules and 23 Solana/Rust pattern rules, with automatic chain detection across 7 EVM networks and Solana, and a Random Forest model ranking findings by exploitability (88% accuracy).
- Shipped it three ways from one codebase — PyPI CLI, GitHub Marketplace Action, and a Next.js/FastAPI web app — hardened with ES256/JWKS token verification, Postgres row-level security, and a blocking CI gate that fails builds on CRITICAL findings.

Self-Healing Edge/IoT Fleet Platform

- Engineered a Java 21 self-healing edge/IoT platform across 5 Maven modules and 279 tests, simulating 50 MQTT devices monitored by a gateway with dual-path failure detection via broker Last Will and heartbeat timeout, streaming to Kafka and persisting to a time-series store;
- Automated recovery through a 300-line Kubernetes operator that replaces dead pods idempotently using an API-server-enforced SHA-256 recovery id, measured at 1332 ms median MTTR across 20 samples at a 20/20 success rate.

Deep Packet Inspection Platform

- Engineered a multi-threaded C++17 deep packet inspection engine using 5-tuple hash routing for flow affinity, parsing Ethernet/IP/TCP/UDP and extracting TLS SNI, HTTP Host, and DNS metadata to classify traffic by application; hardened the parsers against untrusted input with a fuzzing harness validating 85,000+ malformed packets under AddressSanitizer, UBSan, and ThreadSanitizer.
- Built a FastAPI anomaly detection service (scikit-learn Isolation Forest) with percentile-calibrated risk scoring trained on live network captures, served via a Node.js/Express control plane to a Next.js WebSocket dashboard; containerized with Docker Compose and gated by a five-stage GitHub Actions CI pipeline.

Aegis Protocol — Modular Blockchain & Web3 Infrastructure Platform

- Architected a modular DeFi protocol spanning Arbitrum and Solana (vault, oracle network, DAO governance, zk privacy) in Solidity, Rust/Anchor, Go and Noir. Built cross-chain governance via Wormhole, where proposals passed on Arbitrum execute on Solana behind a timelock, a program allowlist and caps on measured treasury outflow.
- Secured and verified the system with UUPS-upgradeable contracts checked for storage-layout safety, a chain-agnostic Go event indexer that is idempotent and safe against chain reorgs, and in-browser zk proofs. Covered it with 900+ automated tests, including 45 end-to-end tests against real local chains, mutation testing and Slither, all enforced through per-layer GitHub Actions CI.

EDUCATION

M.Tech (Integrated) in Software Engineering — VIT Vellore

Expected 2028

CERTIFICATIONS

- IBM DevOps Specialization — DevOps, Agile, and Design Thinking
- 100xDevs Cohort — Web Development, Blockchain, DevOps
- IBM Full Stack Software Developer Specialization — Coursera

LEADERSHIP

Chairman — VIT Blockchain Community

- Organized Web3 workshops, hackathons, and smart contract development programs across the Ethereum and Solana ecosystems